

Tunnel Setup Guide

Connecting Iranian Server to Foreign Server

Protocols: Xray VLESS+Reality & Hysteria2

Feature	VLESS+Reality	Hysteria2
Stealth	Excellent	Good
Speed	High (TCP+BBR)	Very High (QUIC/UDP)
Detection Resistance	Very High	High
Protocol	TCP + TLS 1.3	UDP + QUIC
IP Hiding	Full	Full
Best For	Stealth & reliability	Speed & streaming

1. Overview

This guide provides a complete, production-ready setup for establishing a secure tunnel between an Iranian server and a foreign server. The two recommended protocols, Xray VLESS+Reality and Hysteria2, represent the current state-of-the-art in anti-censorship technology. Both protocols are specifically designed to evade Deep Packet Inspection (DPI) and traffic analysis, which are commonly used for internet censorship and surveillance. The tunnel ensures that your real IP address remains completely hidden from any external observer, while providing high-speed, stable connectivity that can handle everything from web browsing to video streaming and large file transfers.

The VLESS+Reality protocol is the evolution of the V2Ray ecosystem, leveraging TLS 1.3 and a novel key exchange mechanism called Reality. Unlike previous approaches like VLESS+XTLS or Trojan, Reality does not require a valid domain name or TLS certificate on the server. Instead, it transparently proxies the TLS handshake of a real, major website (such as microsoft.com or apple.com), making the traffic indistinguishable from a genuine HTTPS connection to that website. This makes it extremely resistant to active probing and fingerprint-based detection, which are the primary methods used by censors to identify and block tunnel traffic.

Hysteria2 is built on top of the QUIC protocol (UDP-based), which provides significantly better performance on lossy or throttled networks compared to TCP-based protocols. In many regions where TCP traffic is subject to aggressive throttling or QoS (Quality of Service) policies, UDP-based protocols can achieve dramatically higher throughput. Hysteria2 also includes built-in masquerading capabilities, where the server can pretend to serve a legitimate website when probed by censors, and uses authentication to prevent unauthorized access.

2. Architecture

The tunnel architecture follows a client-server model. The foreign server (located outside Iran) runs the server component that listens for incoming tunnel connections. The Iranian server runs the client component, which establishes an encrypted connection to the foreign server and creates local proxy endpoints (SOCKS5 and HTTP). Applications on the Iranian server (or on the local network) can then route their traffic through these proxy endpoints, effectively tunneling all traffic through the encrypted channel to the foreign server.

2.1 VLESS+Reality Architecture

In the VLESS+Reality setup, the Xray server on the foreign machine listens on a port (typically 443) with the Reality security layer enabled. When a legitimate Xray client connects, the Reality protocol performs a special TLS handshake that is cryptographically indistinguishable from a normal TLS 1.3 connection to the target website specified in the SNI field. If an active prober (such as a censor) attempts to connect, the server will transparently proxy the connection to the real target

website, returning a perfectly legitimate TLS certificate and response. This makes the server effectively invisible to active probing attacks, which is one of the most powerful detection methods available to censors.

Component	Iranian Server	Foreign Server
Xray Role	Client	Server
Listening Port	10808 (SOCKS5) / 10809 (HTTP)	443 (VLESS+Reality)
Protocol	VLESS + Reality + XTLS-Vision	VLESS + Reality + XTLS-Vision
Transport	TCP	TCP
Encryption	Reality (TLS 1.3)	Reality (TLS 1.3)

2.2 Hysteria2 Architecture

Hysteria2 operates over QUIC (Quick UDP Internet Connections), which is a UDP-based transport protocol developed by Google and now an IETF standard. Unlike TCP, QUIC supports multiplexed streams without head-of-line blocking, meaning that if one stream experiences packet loss, other streams continue unaffected. Hysteria2 also includes a sophisticated congestion control algorithm called Brutal, which is specifically designed for high-bandwidth, high-latency connections. This allows Hysteria2 to achieve significantly higher throughput than traditional TCP-based protocols, especially on networks with high packet loss or latency.

The Hysteria2 server also implements masquerading: when an unauthorized client or prober connects, the server responds as a reverse proxy to a configured website (e.g., bing.com), serving genuine HTTPS content. This makes the server appear as a normal web server to any external observer, adding another layer of protection against detection and blocking.

Component	Iranian Server	Foreign Server
Hysteria2 Role	Client	Server
Listening Port	10808 (SOCKS5) / 10809 (HTTP)	8443 (UDP)
Protocol	Hysteria2 / QUIC	Hysteria2 / QUIC
Transport	UDP	UDP
Encryption	TLS 1.3 + Auth	TLS 1.3 + Auth
Congestion Control	Brutal	Brutal

3. Prerequisites

Before setting up the tunnel, ensure you have the following prerequisites in place. Both servers need to be properly prepared with the necessary software and network configuration to ensure a smooth installation process and optimal performance. Taking the time to properly prepare your servers will save significant troubleshooting effort later.

3.1 Server Requirements

Requirement	Foreign Server	Iranian Server
OS	Ubuntu 20.04+ / Debian 11+ / CentOS 8+	Ubuntu 20.04+ / Debian 11+ / CentOS 8+
Root Access	Required	Required
RAM	512 MB minimum	256 MB minimum
Bandwidth	Unmetered preferred	Any
Open Ports	443/TCP or 8443/UDP	Outbound 443/TCP, 8443/UDP

3.2 Recommended Foreign Server Locations

The physical location of your foreign server significantly impacts tunnel performance. Choose a location that is geographically close to Iran to minimize latency. Servers in Turkey, Germany, Netherlands, or France typically provide good balance between latency and censorship resilience. Avoid servers in countries with known surveillance agreements or data-sharing arrangements with your region. The following table shows approximate latency from Iran to various locations, based on typical network routing patterns.

Location	Typical Latency	Notes
Turkey (Istanbul)	20-40 ms	Closest, but some ISP throttling
Germany (Frankfurt)	60-90 ms	Good connectivity, reliable
Netherlands (Amsterdam)	70-100 ms	Popular for privacy services
France (Paris)	65-95 ms	Good routing from Middle East
Finland (Helsinki)	80-110 ms	Strong privacy laws

4. Foreign Server Setup

The foreign server setup is the first and most critical step. You must configure the server component on your foreign server before setting up the client on the Iranian server. The provided automation script handles the entire installation process, including downloading and installing the necessary software, generating cryptographic keys, creating configuration files, opening firewall ports, and starting the services. The entire process is fully automated and typically completes in under 5 minutes.

4.1 Quick Setup with Automation Script

The file **tunnel-server-setup.sh** is a fully automated script that handles the complete server-side setup. Simply transfer it to your foreign server and run it as root. The script will guide you through a menu where you can choose which protocol(s) to install, and it automatically generates all necessary keys, configurations, and client connection credentials.

Step-by-step instructions:

- 1. Upload the script to your foreign server:

```
scp tunnel-server-setup.sh root@YOUR_FOREIGN_IP:/root/
```

- 2. SSH into your foreign server and run:

```
chmod +x tunnel-server-setup.sh && sudo ./tunnel-server-setup.sh
```

- 3. Select protocol (1 for VLESS+Reality, 2 for Hysteria2, 3 for both)
- 4. The script will output client connection links - save these securely

Note: The script automatically enables BBR congestion control for TCP optimization and increases UDP buffer sizes for Hysteria2 performance. It also opens the necessary firewall ports for both UFW and firewalld, ensuring compatibility with most server configurations.

4.2 VLESS+Reality Server Configuration

If you prefer to set up manually rather than using the automation script, the following steps describe the VLESS+Reality server configuration in detail. The key components are: the VLESS UUID (a unique identifier for authentication), the x25519 key pair (for the Reality key exchange), the Short ID (an additional authentication parameter), and the SNI/dest domain (the website that the server will masquerade as).

Key Configuration Parameters:

Parameter	Description	Example
UUID	Unique client identifier for authentication	a1b2c3d4-e5f6-...
Private Key	x25519 private key (server only, never share)	Generated by xray x25519
Public Key	x25519 public key (shared with client)	Generated by xray x25519

Short ID	Additional auth parameter, hex encoded	a1b2c3d4e5f6a7b8
SNI / Dest	Real website for Reality masquerading	www.microsoft.com
Flow	XTLS flow control for splicing	xtls-rprx-vision
Fingerprint	TLS client fingerprint (uTLS)	chrome

When choosing the SNI/dest domain, select a major website that supports TLS 1.3 and HTTP/2. Good choices include www.microsoft.com, www.amazon.com, www.apple.com, dl.google.com, or www.tencent.com. The website should be one that is not blocked in the user's region and is unlikely to be blocked in the future. The Reality protocol will use this website's real TLS certificate during the handshake, making the connection appear as a legitimate HTTPS connection to that website.

4.3 Hysteria2 Server Configuration

Hysteria2 server configuration is simpler than VLESS+Reality since it does not require the complex Reality key exchange. The main components are: a password for authentication, a TLS certificate (self-signed is acceptable), and a masquerade URL. The server listens on a UDP port and uses the QUIC protocol for all communication. The Brutal congestion control algorithm is enabled by default, which allows Hysteria2 to achieve significantly higher throughput than traditional TCP-based protocols, especially on networks with high latency or packet loss.

Parameter	Description	Example
Password	Authentication password for clients	Auto-generated (base64)
Port	UDP port to listen on	8443
TLS Cert	Self-signed or custom certificate	/etc/hysteria/server.crt
TLS Key	Private key for the certificate	/etc/hysteria/server.key
Masquerade	Website to proxy for probers	https://www.bing.com
Bandwidth	Server bandwidth limit for Brutal CC	500 mbps up/down

5. Iranian Server (Client) Setup

After the foreign server is configured and running, set up the client on the Iranian server. The client script (**tunnel-client-iran.sh**) handles the complete client-side installation, including installing the Xray or Hysteria2 client software, configuring the connection to the foreign server, setting up local SOCKS5 and HTTP proxy endpoints, and optionally configuring system-wide or transparent proxying so all traffic from the server routes through the tunnel.

5.1 Quick Setup with Automation Script

Transfer the client script to your Iranian server and run it. The script will prompt you for the connection credentials that were generated on the foreign server during the server setup. These credentials are saved in `/root/xray-client-info.txt` (for VLESS+Reality) or `/root/hysteria2-client-info.txt` (for Hysteria2) on the foreign server.

- 1. Upload the script to your Iranian server:

```
scp tunnel-client-iran.sh root@YOUR_IRAN_IP:/root/
```

- 2. SSH into your Iranian server and run:

```
chmod +x tunnel-client-iran.sh && sudo ./tunnel-client-iran.sh
```

- 3. Enter the foreign server IP and credentials when prompted
- 4. Optionally enable system-wide proxy and transparent proxy

5.2 Local Proxy Endpoints

After the client is configured, two local proxy endpoints are created on the Iranian server. These endpoints allow any application on the server (or on the local network, if the server acts as a gateway) to route its traffic through the tunnel. The SOCKS5 proxy supports both TCP and UDP traffic (via the socks5h protocol which performs DNS resolution remotely), while the HTTP proxy handles standard HTTP/HTTPS traffic.

Proxy Type	Address	Usage
SOCKS5	127.0.0.1:10808	<code>curl -x socks5h://127.0.0.1:10808 https://example.com</code>
HTTP	127.0.0.1:10809	<code>curl -x http://127.0.0.1:10809 https://example.com</code>

5.3 System-Wide Proxy

The client script optionally configures system-wide proxy environment variables. When enabled, all HTTP and HTTPS traffic from command-line tools (curl, wget, apt, yum, etc.) will automatically route through the tunnel without requiring per-application configuration. This is achieved by setting the `http_proxy`, `https_proxy`, `HTTP_PROXY`, and `HTTPS_PROXY` environment variables in `/etc/profile.d/tunnel-proxy.sh`. The variables also configure `no_proxy` for local and private network addresses to prevent routing internal traffic through the tunnel.

5.4 Transparent Proxy (Advanced)

For scenarios where you need ALL traffic from the server to route through the tunnel (not just applications that support proxy configuration), the client script can set up transparent proxying using iptables. This uses the REDIRECT target to transparently forward all outgoing TCP

connections through the Xray or Hysteria2 tunnel. The transparent proxy setup automatically excludes private IP ranges and the foreign server IP to prevent routing loops. This is the most seamless option but requires careful configuration to avoid locking yourself out of the server.

Warning: Transparent proxy is an advanced feature. If misconfigured, it can lock you out of your server by routing your SSH connection through the tunnel. The script automatically excludes the foreign server IP to prevent this, but you should always test on a non-critical server first and ensure you have an alternative access method (such as a console/KVM interface) available.

6. Performance Optimization

Optimizing your tunnel for maximum performance involves tuning both system-level parameters and protocol-specific settings. The automation scripts apply several optimizations automatically, but understanding these settings will help you fine-tune performance for your specific network conditions. The most impactful optimizations are BBR congestion control for TCP-based protocols and UDP buffer sizing for QUIC-based protocols.

6.1 BBR Congestion Control (VLESS+Reality)

BBR (Bottleneck Bandwidth and Round-trip propagation time) is a congestion control algorithm developed by Google that significantly outperforms the default Cubic algorithm on high-latency, high-bandwidth connections. BBR estimates the available bandwidth and minimum RTT to determine the optimal sending rate, rather than relying on packet loss as a congestion signal. This is particularly important for tunnel connections, which typically have higher latency than direct connections. The automation script enables BBR automatically by adding the appropriate sysctl parameters to `/etc/sysctl.conf`.

To verify BBR is enabled, run: `sysctl net.ipv4.tcp_congestion_control`

6.2 UDP Buffer Optimization (Hysteria2)

Hysteria2 operates over UDP, and the default Linux UDP buffer sizes are too small for high-throughput tunneling. The automation script increases these buffer sizes to 16 MB for both send and receive, and sets the per-socket defaults to 1 MB. These values allow Hysteria2 to maintain high throughput even on connections with significant bandwidth-delay product. The `netdev_max_backlog` parameter is also increased to 65536 to prevent packet drops during traffic bursts.

6.3 QUIC Parameters (Hysteria2)

The Hysteria2 configuration includes several QUIC-level parameters that can be tuned for optimal performance. The stream and connection receive windows control how much data can be in-flight at any time. Setting these to 8 MB and 20 MB respectively ensures that the connection can fully

utilize high-bandwidth links. The `maxIdleTimeout` is set to 60 seconds to prevent premature connection drops during periods of inactivity. The `disablePathMTUDiscovery` option should be left disabled (false) to allow QUIC to automatically determine the optimal packet size for the connection path.

7. Security Best Practices

While the VLESS+Reality and Hysteria2 protocols provide strong protection against traffic analysis and IP identification, additional security measures should be taken to harden your servers and prevent information leaks. Security is a multi-layered concern, and the tunnel is only as secure as the weakest link in the chain. The following practices will help you maintain a robust security posture.

7.1 DNS Leak Prevention

One of the most common information leaks is DNS queries bypassing the tunnel. By default, applications may use the system DNS resolver, which sends queries directly to your ISP's DNS servers, revealing the domains you are accessing. The Xray client configuration includes sniffing enabled on the inbound proxy, which intercepts DNS queries and routes them through the tunnel. For additional protection, configure your system DNS to use a secure resolver such as 1.1.1.1 or 8.8.8.8, and ensure that all DNS traffic is routed through the tunnel by using the SOCKS5 proxy with remote DNS resolution (`socks5h://`).

7.2 Server Hardening

Both servers should be hardened using standard Linux security practices. This includes: disabling password authentication and using SSH keys only, disabling root login via SSH where possible, keeping the system updated with security patches, using a firewall to allow only necessary ports, installing and configuring fail2ban to prevent brute-force attacks, and regularly reviewing system logs for suspicious activity. The foreign server should also have any unnecessary services disabled to reduce the attack surface.

7.3 Credential Management

The tunnel credentials (UUID, passwords, keys) should be treated as sensitive information and protected accordingly. Never share credentials over unencrypted channels. Store them in encrypted files or password managers. Rotate credentials periodically, especially if you suspect they may have been compromised. The automation scripts save credentials to `/root/` on the respective servers with restrictive file permissions (only root can read), but you should also maintain a secure offline backup of these credentials.

7.4 Monitoring and Maintenance

Regular monitoring is essential to ensure the tunnel remains operational and secure. Check service status periodically using `systemctl status xray` or `systemctl status hysteria-server`. Monitor connection logs for any unusual activity or connection attempts from unknown IPs. Set up automated alerts for service failures. Keep the Xray and Hysteria2 software updated to the latest versions to benefit from security patches and performance improvements. The Xray project releases updates frequently, and staying current is important for maintaining detection resistance.

8. Troubleshooting

If the tunnel is not working as expected, the following troubleshooting steps will help you identify and resolve common issues. Start with the basic connectivity checks and work your way through the more specific protocol-level diagnostics.

Issue	Possible Cause	Solution
Connection timeout	Firewall blocking port	Check firewall rules on both servers; verify port is open
Connection refused	Service not running	Check systemctl status; restart service
TLS handshake failure	Wrong SNI or credentials	Verify SNI matches server config; check UUID/keys
Slow speeds	BBR not enabled, buffer too small	Enable BBR; increase UDP buffers for Hysteria2
DNS leaks	System DNS not through tunnel	Use socks5h:// or configure DNS in Xray routing
Service crashes	Config syntax error	Validate JSON config; check journalctl logs
UDP not working	ISP blocking UDP	Switch to VLESS+Reality (TCP-based)

8.1 Diagnostic Commands

The following commands are useful for diagnosing tunnel connectivity and performance issues. Run these on the Iranian server to verify that the tunnel is working correctly and identify any potential problems.

```
# Check if the proxy is responding
curl -x socks5h://127.0.0.1:10808 -s https://api.ipify.org
```

```
# Test with HTTP proxy
curl -x http://127.0.0.1:10809 -s https://api.ipify.org

# Compare with direct connection
curl -s https://api.ipify.org

# Check service status
systemctl status xray # or hysteria-client

# View recent logs
journalctl -u xray -n 50 --no-pager
```

9. Recommended Client Applications

In addition to the server-to-server tunnel setup described in this guide, you can also use the generated connection links with various client applications on desktop and mobile devices. This allows individual devices to connect directly to the foreign server without requiring the Iranian server as an intermediary. The VLESS link and Hysteria2 link generated by the server setup script can be imported into any compatible client application.

Application	Platform	Protocols	Notes
v2rayN	Windows	VLESS+Reality	Most popular Windows client
Nekobox	Android	Both	Supports both protocols
V2rayU	macOS	VLESS+Reality	Good macOS option
v2rayA	Linux (Web UI)	Both	Web-based management
Clash Meta	All platforms	Both	Advanced routing rules
Shadowrocket	iOS	Both	Paid app, reliable
FoXray	iOS	VLESS+Reality	Free alternative

10. Quick Reference

This section provides a condensed summary of the key steps, commands, and configuration details for quick reference during setup and maintenance operations.

10.1 Setup Summary

Step	Action	Server
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1	Run tunnel-server-setup.sh	Foreign
2	Save generated credentials (VLESS link / Hysteria2 link)	Foreign
3	Run tunnel-client-iran.sh	Iranian
4	Enter foreign server IP and credentials	Iranian
5	Optionally enable system-wide / transparent proxy	Iranian
6	Verify with: curl -x socks5h://127.0.0.1:10808 https://api.ipify.org	Iranian

10.2 Key File Locations

File	Path	Server
Xray Server Config	/usr/local/etc/xray/config.json	Foreign
Xray Client Info	/root/xray-client-info.txt	Foreign
Hysteria2 Server Config	/etc/hysteria/config.yaml	Foreign
Hysteria2 Client Info	/root/hysteria2-client-info.txt	Foreign
Hysteria2 Client Config	/etc/hysteria/client.yaml	Iranian
Proxy Env Variables	/etc/profile.d/tunnel-proxy.sh	Iranian

10.3 Service Management Commands

Action	Xray Command	Hysteria2 Command
Start	systemctl start xray	systemctl start hysteria-server
Stop	systemctl stop xray	systemctl stop hysteria-server
Restart	systemctl restart xray	systemctl restart hysteria-server
Status	systemctl status xray	systemctl status hysteria-server
Logs	journalctl -u xray -f	journalctl -u hysteria-server -f